

Rajan Kumar Gupta

Electronics Engineer | Backend & Full-Stack Development | DevOps

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PROFILE SUMMARY

Electronics, Communication and Information Engineering graduate and backend/full-stack developer (MERN, REST APIs, PostgreSQL/Supabase) with hands-on DevOps experience in Docker, GitHub Actions CI/CD, Kubernetes (k3s), Helm, Ansible, and Prometheus/Grafana on Linux (AlmaLinux). Completed formal DevOps training at Tech Axis (2026) and built a full-stack Theatre Booking System with authentication, payments, and admin dashboards. Experienced technical mentor seeking a DevOps/cloud role.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, C/C++

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, Django, Flask

Databases: PostgreSQL, Supabase, MongoDB, MySQL, SQL Server, SQLite

DevOps & Infrastructure: Linux (AlmaLinux), Git, GitHub Actions (CI/CD, self-hosted runner), Docker, Docker Compose, Kubernetes, k3s, Helm, Ansible, Prometheus, Grafana, Traefik / Kubernetes Ingress, Kubernetes HPA, Kubernetes PVC (persistent storage)

Cloud: AWS (fundamentals)

Tools & Platforms: Git, GitHub, Postman, Render, Vercel, Netlify

Embedded & IoT: Raspberry Pi, Arduino, ESP8266/ESP32, AD8232 ECG Sensor, RS485, RS232, UART

Others: TensorFlow, OpenCV

WORK EXPERIENCE

MERN Stack Intern – Digital Pathshala Pvt. Ltd., Itahari, Sunsari | May 5 – June 9, 2025

- Developed REST APIs for a Theatre Booking System using Node.js, Express, and Supabase (PostgreSQL).
- Implemented JWT authentication, role-based access, and Khalti payment verification.
- Built real-time seat booking, cancellation, and overlap validation logic.
- Designed optimized PostgreSQL tables and improved API response performance.
- Integrated Supabase Storage for uploading show images and user assets.

DEVOPS PROJECTS

Hands-on projects built and run in a personal lab environment (not professional production infrastructure).

WordPress DevOps CI/CD Pipeline | github.com/rajanshah23/wordpress-docker-cicd

AlmaLinux | Docker | Docker Compose | GitHub Actions | Kubernetes/k3s | Helm | Ansible | Prometheus | Grafana

- Built a Linux DevOps environment on an AlmaLinux 9 VM: static IP networking, non-root sudo user, and an XFS secondary disk with a persistent /etc/fstab mount.
- Containerized WordPress and MySQL with Docker Compose, using named volumes and a custom Docker network.
- Configured a self-hosted GitHub Actions runner as a systemd service; the workflow deploys the stack on every push to main, with database credentials in GitHub Secrets.
- Deployed on k3s with kubectl and manifests (Namespace, Secret, ConfigMap, PV/PVC, Deployments, Services, Traefik Ingress) and a CPU-based HPA scaling WordPress from 1 to 3 replicas via Metrics Server.
- Created a reusable Helm chart and installed kube-prometheus-stack to monitor the cluster with Prometheus and Grafana.
- Wrote an Ansible playbook with roles (common, Docker, k3s, Helm, WordPress, monitoring) to automate setup on a fresh VM.

DevOps Lab Pipeline | github.com/rajanshah23/devops-lab-pipeline

Docker | Docker Compose | Prometheus | Grafana | Node.js | Jest | Supertest

- Containerized a Node.js 20 / Express REST API with a multi-stage Alpine Dockerfile, non-root runtime user, and healthcheck; image published to Docker Hub.
- Built a Docker Compose stack running the API, Prometheus, and Grafana on a shared network with health-gated startup, runnable with one command.
- Implemented /health, /ready, and /metrics endpoints with HTTP request counts, request-duration histograms, and Node.js runtime metrics for Prometheus.
- Provisioned a Grafana dashboard as code: request rate, 5xx error rate, p95 latency, and Node heap usage.

- Added automated tests with Jest and Supertest, plus ESLint configuration.

WEB DEVELOPMENT PROJECTS

- **Theatre Booking System:** Full-stack app using React.js, TypeScript, Tailwind CSS, Node.js, Express, and PostgreSQL with Supabase. Features JWT auth, seat booking, Khalti payment, review system, and image storage via Supabase. (github.com/rajanishah23/TheatreBooking-System)
- **IncorpoFlow: Company Incorporation Tool:** Full-stack multi-step company registration app with draft persistence and admin view, built with React (Vite), Node.js, Express, PostgreSQL (Sequelize ORM), and Docker. Drafts save instantly across browser refreshes; RESTful API with full CRUD and cascading deletes. Entire stack (PostgreSQL, backend, Nginx-served frontend) containerized with Docker Compose for one-command setup. Admin dashboard lists all companies with their shareholders.
- **Online Shopping Center:** Full-stack MERN app with product browsing, search and filter, cart management, order processing, and secure user authentication.
- **Currency Converter Web App:** Responsive app for real-time currency conversion using REST APIs, HTML, CSS, and JavaScript.
- **Paste App:** Lightweight text snippet manager with React, Redux Toolkit, and LocalStorage, with a Tailwind CSS UI and React Hot Toast notifications.

TRAINING & CERTIFICATIONS

- DevOps Training – Certificate of Completion, Tech Axis Pvt. Ltd., April 28 – August 7, 2026 (Certificate No. DO/26/114/14067; issued 16 Aug 2026). Verify: techaxis.com.np/certification

EDUCATION

Bachelor in Electronics, Communication and Information Engineering: Paschimanchal Campus, Tribhuvan University | Pokhara, Nepal, 2021 – 2025

Higher Secondary Education: Kalika Manavgyan Secondary School | Butwal, Nepal, 2018 – 2020

PUBLICATIONS

[1] A. Laudari Bhat, B. KC Khatri, G. Rijal, R. K. Gupta, and S. Adhikari, "Obstacle Detection for Visually Impaired," Proceedings of IOE Graduate Conference, Vol. 16, pp. 1913–1919, April 2025 (Published June 2026). Peer-reviewed. ISSN: 2350-8914 (Online), 2350-8906 (Print).

TEACHING, MENTORING & LEADERSHIP

Technical Mentor – Paschimanchal Campus | 2023 – 2024

- Mentored 15+ junior students in full-stack web development, guiding them from fundamentals to project completion.
- Conducted weekly hands-on sessions covering React.js, Node.js, and database design.
- Organized and led soldering and PCB design workshops for 30+ participants.
- Provided one-on-one guidance for student projects and academic challenges, including embedded systems.
- Led project teams in academic and extracurricular projects; attended tech events and bootcamps to stay current with industry trends.

ACADEMIC & HARDWARE PROJECTS

- **ECG Monitoring System:** Designed and implemented an ECG monitoring tool for biomedical applications with live visualization. (Paschimanchal Campus | 2023 – 2024)
- **Obstacle Detection for Visually Impaired:** Assistive system using ultrasonic sensors and haptic feedback for real-time guidance; published at IOE Graduate Conference, April 2025 (lnkd.in/g8TuWbnm). (Paschimanchal Campus | 2024 – 2025)